# Rational Functions and Asymptotes

## Finding asymptotes and intercepts

1 For $f(x) = \frac{x+2}{x-3}$, find:

**a** the vertical asymptote

**b** the horizontal asymptote

**c** the $x$-intercept

**d** the $y$-intercept

2 Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.

3 Consider $f(x) = \frac{x^2-9}{x-3}$.

**a** Simplify $f$ and identify the feature at $x = 3$.

**b** Does the graph have a vertical asymptote?

4 Find the slant (oblique) asymptote of $f(x) = \frac{x^2+1}{x-1}$.

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## Behaviour near asymptotes

5 For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:

**a** $x \to 3^+$

**b** $x \to 3^-$

**c** $x \to \infty$

6 Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.

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## Classifying end behaviour from degree

7 Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:

**a** $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2)

**b** $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2)

**c** $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2)

8 Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.

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## Holes versus asymptotes

9 For $f(x) = \frac{(x-2)(x+5)}{(x-2)(x-1)}$, identify the hole and the vertical asymptote.

10 Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.